

Notes: Ryan (Econometrica, 2012)

The Costs of Environmental Regulation in a Concentrated Industry

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1 Big picture

- **Question.** What are the product-market and welfare costs of environmental regulation when the regulation changes industry entry, exit, investment, and market structure rather than only imposing engineering compliance costs?
- **Setting.** The paper studies the U.S. Portland cement industry and the 1990 Amendments to the Clean Air Act. Cement is a pollution-intensive, capital-intensive, locally segmented, concentrated industry.
- **Core critique of standard policy analysis.** A standard engineering cost estimate measures compliance equipment and monitoring expenditures. In a concentrated industry, this misses the dynamic effects on entry, capacity investment, exit, prices, and consumer surplus.
- **Model.** The paper estimates a dynamic oligopoly model in the spirit of Maskin–Tirole and Ericson–Pakes. Firms choose entry, exit, capacity investment, and product-market quantities in regional markets.
- **Estimation.** The paper uses the two-step method of Bajari, Benkard, and Levin. First, it estimates demand, product-market profits, and firms’ policy functions. Second, it recovers structural distributions of entry costs, investment costs, divestment costs, scrap values, and exit costs.
- **Main empirical result.** The 1990 Amendments increased sunk entry costs substantially. This reduced entry, changed market structure, and shifted surplus away from consumers.
- **Main counterfactual result.** Product-market surplus falls by roughly hundreds of millions to several billion dollars depending on the market state and policy experiment. Consumer surplus losses are large because higher entry costs reduce competition and raise prices.
- **Sign reversal result.** A static analysis can get the sign of incumbent welfare effects wrong. Incumbents may gain from regulation because higher entry costs protect them from future competition.
- **Takeaway.** Environmental regulation can impose important indirect costs through industrial organization. Measuring only plant-level compliance costs misses the market-structure channel.

2 Incremental contribution of the paper

2.1 Contribution relative to environmental regulation cost studies

- Traditional regulatory cost analysis often focuses on engineering costs: what firms must spend to install pollution-abatement equipment or satisfy monitoring requirements.
- Ryan’s paper contributes by treating regulation as a shock to the dynamic cost structure of an industry. The regulation affects firms’ incentives to enter, stay, exit, and adjust capacity.
- This moves the analysis from direct compliance costs to equilibrium welfare costs. The relevant welfare effect includes induced price changes, concentration changes, and changes in producer and consumer surplus.

Interpretation: The incremental contribution is to show that environmental policy costs are not only technological costs. In concentrated industries, regulation changes market structure, and this can dominate the welfare accounting.

2.2 Contribution relative to static IO studies

- Static demand and supply models can estimate price and quantity effects conditional on a fixed number of firms.
- This paper allows the number and size of firms to change endogenously through entry, exit, and investment.

- The static model misses that firms may enter less or invest differently when the policy raises sunk costs.

Interpretation: The paper’s central mechanism is dynamic: the regulation changes future competitive structure. Static models cannot capture the consumer surplus penalty from reduced entry.

2.3 Contribution relative to dynamic oligopoly estimation

- The paper applies the Bajari, Benkard, and Levin two-step estimator to a policy question where dynamic market structure is central.
- It recovers not only variable production costs but also distributions of sunk entry, capacity adjustment, divestment, and scrap value parameters.
- It then uses the estimated primitives to solve and simulate counterfactual Markov-perfect equilibria under alternative regulatory cost structures.

Interpretation: Methodologically, the paper demonstrates how dynamic game estimation can be used for environmental policy evaluation.

2.4 Contribution relative to studies of the Portland cement industry

- The Portland cement industry is geographically segmented because cement is heavy and costly to transport.
- It is also concentrated and capacity constrained, making entry and investment central to long-run outcomes.
- The paper builds a detailed regional-market panel using demand, production, and plant capacity data.

Interpretation: The paper’s empirical setting is well suited to dynamic oligopoly analysis because both market power and physical capacity constraints matter.

3 Data and measurement

3.1 Industry background

- Portland cement is the binding input in concrete and is used in construction, roads, buildings, and infrastructure.
- Cement production is energy intensive and produces significant air pollution.
- The industry is geographically localized because transport costs are high relative to product value.
- The industry is highly capital intensive: plants require large kilns, fixed facilities, and long-lived capacity.
- Entry and capacity adjustment are costly and slow, so current market structure reflects past investment and entry choices.

Interpretation: These features make Portland cement a natural environment for studying regulation through dynamic market structure.

3.2 Clean Air Act Amendments

- The 1990 Amendments expanded environmental regulation and changed permitting and certification requirements.
- For cement producers, the Amendments increased compliance burdens and made new entry more costly.
- The paper estimates cost primitives separately before and after 1990 to identify how the regulatory environment changed the industry’s dynamic cost structure.

Interpretation: The paper interprets the post-1990 shift in entry costs as the central regulatory cost shock.

3.3 Market-level demand and supply data

- Demand data include annual market-level quantities and prices for regional cement markets.
- Demand shifters include population and construction permits.
- Cost shifters used as instruments include energy and input prices such as gas, coal, electricity, and skilled wages.
- Market-level data cover 27 regional markets and 517 market-year observations in the demand estimation.

Interpretation: Regional variation in prices, quantities, and demand shifters identifies cement demand and helps recover product-market profits.

3.4 Plant-level production and capacity data

- Plant-level production and capacity data come from the Portland Cement Association’s annual plant information summaries.
- Production and capacity are measured in thousands of tons per year.
- Capacity is interpreted as the amount of cement that could be produced in a year.
- Capacity adjustments are infrequent and often large, consistent with lumpy investment and high fixed costs.

Interpretation: The plant data are crucial because the state variable in the dynamic model is the capacity vector of firms in each market.

4 Model and empirical strategy

4.1 State variables and timing

Each regional market is described by a state vector

$$s_t = (s_{1t}, \dots, s_{Nt}), \quad (1)$$

where s_{it} is firm i ’s capacity in period t and N is the maximum number of potentially active firms.

In each period:

1. incumbents draw exit/scrap shocks and choose whether to exit;
2. potential entrants draw entry and investment costs;
3. incumbents and entrants choose capacity adjustments and entry;
4. active firms compete in quantities;
5. entry, exit, and capacity investments become part of next period’s state.

Interpretation: The timing captures that investment and entry take time to affect capacity, while product-market competition happens with current capacity.

4.2 Demand

The market demand curve is constant elasticity:

$$\ln Q_m = \alpha_{0m} + \alpha_1 \ln P_m, \quad (2)$$

where Q_m is aggregate quantity, P_m is price, and α_{0m} is a market intercept.

Interpretation: The estimated demand elasticity determines how changes in entry and capacity translate into price and consumer surplus effects.

4.3 Production costs

Firm i 's production cost is

$$C_i(q_i; s_i) = \delta_0 + \delta_1 q_i + \delta_2 \mathbf{1}\{q_i > \nu s_i\} (q_i - \nu s_i)^2. \quad (3)$$

- δ_0 is fixed operating cost.
- δ_1 is constant marginal cost.
- The quadratic term captures increasing costs near capacity.
- The threshold ν determines when capacity pressure starts to bind.

Interpretation: The hockey-stick cost curve allows firms to produce near capacity but at increasing marginal cost.

4.4 Capacity adjustment costs

Capacity adjustment cost is

$$\Gamma(x_i; \gamma) = \mathbf{1}\{x_i > 0\}(\gamma_{1i} + \gamma_2 x_i + \gamma_3 x_i^2) + \mathbf{1}\{x_i < 0\}(\gamma_{4i} + \gamma_5 x_i + \gamma_6 x_i^2). \quad (4)$$

- Positive x_i means investment.
- Negative x_i means divestment or capacity reduction.
- Fixed investment and divestment costs are private shocks drawn each period.

Interpretation: This cost structure allows lumpy capacity adjustment and asymmetric investment/divestment costs.

4.5 Entry, exit, and per-period payoff

Entrants pay a sunk entry cost κ_i . Exiting firms receive a scrap value ϕ_i .

The per-period payoff can be written as

$$\pi_i(s, a; \theta, \varepsilon_i) = \pi_i^{prod}(s; \alpha, \delta) - \Gamma(x_i; \gamma) - \mathbf{1}\{entry\}\kappa_i + \mathbf{1}\{exit\}\phi_i. \quad (5)$$

Interpretation: The structural parameters determine how regulation changes market structure: by changing the entry-cost distribution or adjustment-cost distribution, the policy changes future competition.

4.6 Equilibrium

Firms use symmetric anonymous Markov strategies:

$$\sigma_i : (s, \varepsilon_i) \mapsto a_i. \quad (6)$$

The value function for an incumbent combines current product-market profit, investment/divestment choices, and continuation value:

$$V_i(s) = \max_{a_i} \{ \pi_i(s, a_i, a_{-i}; \theta) + \beta E[V_i(s') \mid s, a] \}. \quad (7)$$

Interpretation: The Markov-perfect restriction makes the dynamic game tractable and allows estimation using observed policy functions.

5 Estimation

5.1 Step 1: demand, production, and policy functions

The first step estimates objects that describe observed behavior:

- demand elasticity and market intercepts;
- production cost parameters;
- investment policy functions using flexible basis functions;
- entry and exit policy functions using probit specifications.

The investment policy is summarized by an (S, s) style rule: firms adjust capacity when their current capacity is far enough from a target level. The adjustment band and target are estimated as functions of own capacity, competitors' capacity, population, and post-1990 indicators.

Interpretation: Step 1 describes what firms do at each state, without yet recovering the cost primitives that rationalize those choices.

5.2 Step 2: structural parameters

The second step searches for structural parameters that make observed policies optimal. The BBL minimum-distance estimator compares observed policy values to values from alternative policies. The estimated primitive parameters include:

- investment costs;
- divestment costs;
- fixed investment and divestment cost distributions;
- scrap value distribution;
- entry cost distribution.

Interpretation: This step recovers the dynamic cost structure of the industry, which is the object needed for counterfactual policy analysis.

5.3 Standard errors

Standard errors are computed by subsampling complete market histories. The unit of resampling is a market history rather than an isolated observation.

Interpretation: This reflects the serial dependence induced by dynamic market evolution.

6 Identification and empirical design

6.1 What identifies demand and marginal costs

- Demand uses variation in prices, quantities, population, and construction permits across regional markets and years.
- Input prices such as gas, coal, electricity, and skilled wages are instruments for price.
- Production costs are identified from plant-level quantities and capacity utilization patterns.

Interpretation: Cost shifters help distinguish movement along demand from shifts in demand.

6.2 What identifies adjustment and entry costs

- Observed capacity adjustments identify investment and divestment policy functions.
- Infrequent entry identifies the distribution of sunk entry costs.
- Exit decisions identify scrap values and exit costs.
- Before-versus-after 1990 comparisons identify how the regulatory environment changed structural costs.

Interpretation: The post-1990 shift is interpreted as the impact of the Clean Air Act Amendments on dynamic industry primitives.

6.3 Limitations

- The model assumes independent regional markets and rules out multimarket contact.
- It restricts firm strategies to anonymous symmetric Markov strategies.
- The counterfactual simulations focus on representative market states rather than solving every regional market.
- The welfare analysis measures product-market effects and excludes the health and environmental benefits of cleaner air.

Interpretation: The estimates isolate the industrial-organization costs of regulation, not the full social cost-benefit comparison.

7 Tables and figures

7.1 Tables I and II: industry and data summary

Read:

- Table I summarizes the Portland cement industry from 1980 to 1998.
- Production, imports, consumption, prices, capacity, and capacity per kiln are reported annually.
- Capacity per kiln rises steadily, and imports grow substantially.
- Table II summarizes market-level demand data, plant-level production data, and plant-level investment data.
- The mean market has about 4.75 plants and mean annual quantity of about 2.84 million tons.

Economic message: These tables establish the industry's key facts: capacity is persistent, markets are concentrated, and capacity adjustment is relatively rare.

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TABLE I
CEMENT INDUSTRY SUMMARY STATISTICS^a

Year	Production	Imports	Consumption	Price	Capacity	Capacity per Kiln
1980	68,242	3035	70,173	111.90	89,561	239
1981	65,054	2514	66,092	103.70	93,203	267
1982	57,475	2231	59,572	95.76	89,770	287
1983	63,884	2960	65,838	91.01	92,052	292
1984	70,488	6016	76,186	89.70	91,048	297
1985	70,665	8939	78,836	84.71	88,600	305
1986	71,473	11,201	82,837	81.48	87,341	305
1987	70,940	12,753	84,204	78.07	86,709	314
1988	69,733	14,124	83,851	75.50	86,959	327
1989	70,025	12,697	82,414	72.04	84,515	337
1990	69,954	10,344	80,964	69.02	83,955	345
1991	66,755	6548	71,800	66.37	84,471	352
1992	69,585	4582	76,169	64.25	85,079	357
1993	73,807	5532	79,701	63.58	84,869	363
1994	77,948	9074	86,476	68.06	85,345	364
1995	76,906	10,969	86,003	72.56	86,285	367
1996	79,266	11,565	90,355	73.64	85,687	376
1997	82,582	14,523	96,018	74.60	86,465	383
1998	83,931	19,878	103,457	76.45	87,763	393

^aSummary statistics for the Portland cement industry 1980–1998. The data are from Historical Statistics for Mineral and Materials Commodities in the United States, an online publication of the U.S. Geological Survey. The units on quantities are thousands of metric tons, while prices are denoted in 1998 constant dollars.

Table I: cement industry summary statistics

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TABLE II
SUMMARY STATISTICS^a

Variable	Minimum	Mean	Maximum	Standard Deviation
Market-Level Supply and Demand Data				
Quantity	186	2835.84	10,262	1565.34
Price	36.68	67.46	138.99	13.68
Plants in market	1	4.75	20	1.94
Skilled wage	20.14	31.72	44.34	4.33
Coal price	15.88	26.64	42.33	8.13
Electricity price	4.23	5.68	7.6	1.01
Natural gas price	3.7	6.21	24.3	2.21
Population	689,584	10,224,352	33,145,121	7,416,485
Plant-Level Production Data				
Quantity	177	699	2348	335
Capacity	196	797	2678	386
Plant-Level Investment				
Capacity investment	−728	2.19	1140	77.60

^aDemand data are from annual volumes of the USGS's Mineral Yearbook, 1980–1981 to 1998–1999. There are 517 observations in 27 regional markets. Quantities and capacities are denominated in thousands of tons per year, while price is denoted in dollars per ton. Labor wages are denoted in dollars per hour for skilled manufacturing workers, and taken from County Business Patterns. Population is the total populations of the states covered by a regional market. The units are dollars per ton for coal, dollars per kilowatt hour for electricity, and dollars per thousand cubic feet for gas. All prices are adjusted to 1996 constant dollars. The data on production and capacity are taken from the Portland Cement Association's annual Plant Information Summary, with full coverage from 1980 to 1998.

Table II: demand, production, and investment summary statistics

7.2 Tables III and IV: demand and production

Read:

- Table III estimates cement demand. Price elasticity estimates are negative and precisely estimated in many specifications.
- Log population and construction permits shift demand positively.
- Table IV estimates production costs. Marginal cost is about 31.6, capacity cost is positive, and the capacity-cost threshold implies costs begin to bind at high utilization.
- Average prices, revenues, costs, profits, and margins are also summarized.

Economic message: Demand and production estimates provide the product-market profit function used in the dynamic game.

TABLE IV
PRODUCTION FUNCTION RESULTS^a

Production Function Estimates		
Parameter	Coefficient	Standard Error
Marginal cost (δ_1)	31.58	1.91
Capacity cost (δ_2)	1.239	0.455
Capacity cost threshold ($\bar{p}$)	1.916	0.010
Marginal cost post-1990 shifter	2.41	3.33
Capacity cost post-1990 shifter	-0.0299	0.22
Capacity cost threshold post-1990 shifter	0.0917	0.0801
Prices, Revenues, and Profits		
Variable	Value	Standard Deviation
Price	57.81	16.83
Revenues	39,040	19,523
Costs	22,525	11,051
Profit	16,515	12,244
Margin	39.29 percent	18.21 percent

^aThe binding threshold at which the capacity costs become important is restricted to $[0, 1]$ by estimating a logit probability: $p = \exp(\bar{p}) / (1.0 + \exp(\bar{p}))$. At the estimated value of 1.916, this implies that capacity costs start to bind at an approximately 87 percent utilization rate.

TABLE III
CEMENT DEMAND ESTIMATES^a

	I	II	III	IV	V	VI
Price	-3.21 (0.361)	-1.99 (0.285)	-2.96 (0.378)	-0.294 (0.176)	-2.26 (0.393)	-0.146 (0.127)
Intercept	21.3 (1.52)	10.30 (1.51)	20.38 (1.56)	-3.41 (1.09)	11.6 (2.04)	-6.43 (0.741)
Log population		0.368 (0.0347)		0.840 (0.036)	0.213 (0.074)	0.789 (0.033)
Log permits					0.218 (0.072)	0.332 (0.035)
Market fixed effects	No	No	Yes	Yes	No	Yes

Table III: demand estimates

Table IV: production function results

7.3 Tables V and VIII: productivity, entry, and exit policies

Read:

- Table V estimates productivity functions. Capacity explains production well, and there is little evidence that entrants or exiting firms have sharply different productivity per unit of capacity.
- Table VIII estimates entry and exit policies. Exit probability rises when own capacity is low and competitors' capacity is high. Entry becomes less likely after 1990.
- The post-1990 indicators in entry and exit policies are important for understanding how the Amendments changed market dynamics.

Economic message: The reduced-form policy functions describe how firms change capacity and participation as market structure changes.

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TABLE V
PRODUCTIVITY ESTIMATES^a

Specification	I	II	III	IV	V
Capacity	0.8617 (0.002)	0.8600 (0.002)	0.860 (0.002)	0.860 (0.002)	0.860 (0.002)
Rivals' capacity	-0.007 (0.001)	-0.005 (0.001)	-0.002 (0.001)	-0.003 (0.001)	0.0003 (0.0006)
Firm entered * capacity		0.0009 (0.0027)	0.0002 (0.0027)	0.0112 (0.0064)	0.0103 (0.007)
Firm exited * capacity		-0.0154 (0.0035)	-0.0128 (0.0036)	-0.0173 (0.0078)	-0.0135 (0.008)
Time trend			0.671 (0.130)	0.681 (0.131)	
Entry dummy				-11.66 (6.141)	-11.49 (6.678)
Exit dummy				3.041 (4.810)	0.492 (5.107)
Market fixed effects	Yes	Yes	Yes	Yes	No
Market-time fixed effects	No	No	No	No	Yes
R ²	0.9925	0.9925	0.9926	0.9926	0.9933

^aNumber of observations = 2233.

Table V: productivity estimates

TABLE VIII
ENTRY AND EXIT POLICY RESULTS^a

Specification	I	II	III	IV
Exit Policy				
Own capacity	-0.0015661 (0.000268)	-0.0015795 (0.0002712)		
Competitors' capacity	0.0000456 (0.0000173)	0.0000379 (0.0000249)		
Population		0.0590591 (0.1371835)		
After 1990	-0.5952687 (0.1616594)	-0.606719 (0.1639955)	-0.6328867 (0.157673)	-0.4623664 (0.1910193)
Own capacity per capita			-0.0005645 (0.0001255)	-0.0010199 (0.0002164)
Competitors' capacity per capita			0.0000744 (0.0000286)	0.0002379 (0.0001023)
Constant	-1.000619 (0.1712286)	-1.019208 (0.176476)	-1.664808 (0.1475588)	-1.529715 (0.3526938)
Region fixed effects	No	No	No	Yes
Log-likelihood	-227.21	-227.12	-238.54	-217.38
Entry Policy				
Competitors' capacity	0.0000448 (0.0000365)	-0.0003727 (0.0002351)		
After 1990	-0.6089773 (0.2639545)	-0.8781589 (0.3229502)	-0.602279 (0.2651052)	-1.003239 (0.337589)
Constant	-1.714599 (0.2152315)	-0.454613 (0.7086509)	-1.665322 (0.2642566)	-0.3434765 (0.6624767)
Competitors' capacity per capita			0.000026 (0.000038)	-0.0003633 (0.0001766)
Region fixed effects	No	Yes	No	Yes
Log-likelihood	-70.01	-56.47	-70.491	-55.53
Prob > χ^2	0.0177	0.4516	0.0287	0.3328

^aSample size for exit policy function = 2233; sample size for entry policy function = 414. Capacity is measured in thousands of tons of cement per year. Population is normalized to be measured in tens of millions. Per-capita capacity is measured as thousands of tons per year per tens of millions in population.

Table VIII: entry and exit policy results

7.4 Tables VI and VII: investment policy

Read:

- Table VI estimates the adjustment band size in the (S, s) investment rule.
- Table VII estimates the investment target.
- Both tables use flexible b-splines in competitors' capacity, own capacity, and population.
- The high adjusted R^2 values indicate that the policy functions fit observed capacity adjustments well.

Economic message: Investment policy functions are key to the dynamic model because capacity is the state variable that determines future competition.

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TABLE VI
INVESTMENT POLICY FUNCTION RESULTS: ADJUSTMENT BAND SIZE^a

Specification	I	II	III	IV
Sum competitors' capacity b-spline 1	6.31 (0.973)	7.18 (5.42)	3.29 (0.827)	5.74 (1.14)
Sum competitors' capacity b-spline 2	6.51 (0.930)	7.16 (5.43)	2.04 (0.953)	4.53 (1.37)
Sum competitors' capacity b-spline 3	5.66 (0.910)	6.41 (5.40)	3.57 (0.888)	4.89 (1.28)
Sum competitors' capacity b-spline 4	6.98 (0.960)	7.85 (5.46)	2.05 (0.978)	4.05 (1.36)
Sum competitors' capacity b-spline 5	5.77 (0.939)	6.72 (5.33)	2.91 (0.994)	5.40 (1.47)
Sum competitors' capacity b-spline 6	7.3 (0.944)	8.15 (5.67)	2.11 (1.15)	4.23 (1.50)
Own capacity b-spline 1	-3.79 (0.923)	-3.97 (0.925)	0.374 (0.880)	-2.71 (1.22)
Own capacity b-spline 2	-3.3 (0.893)	-3.37 (0.894)	0.720 (0.902)	-0.754 (1.22)
Own capacity b-spline 3	-2.3 (0.967)	-2.51 (0.969)	1.06 (1.04)	-0.325 (1.28)
Own capacity b-spline 4	-1.72 (0.943)	-1.76 (0.952)	1.87 (1.27)	-0.149 (1.60)
Own capacity b-spline 5	-2.63 (1.35)	-2.66 (1.35)	2.05 (2.25)	3.32 (2.02)
Population b-spline 1		-5.11 (6.78)		
Population b-spline 2		0.886 (5.16)		
Population b-spline 3		-1.39 (5.52)		
Population b-spline 4		-0.008 (5.06)		
Population b-spline 5		-1.60 (6.69)		
Capacity is per capita	No	No	Yes	Yes
Region fixed effects	No	No	No	Yes
Adjusted R^2	0.8952	0.8955	0.8816	0.8946
Band σ^2	1.40	1.40	1.56	1.41

^aDependent variable is the natural log of the change in capacity. Number of capacity changes = 774. Parameters estimated using ordinary least squares (OLS). Capacity is measured in thousands of tons per year. Population is denominated in tens of millions.

Table VI: adjustment band size

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TABLE VII
INVESTMENT POLICY FUNCTION RESULTS: INVESTMENT TARGET^a

	I	II	III	IV
Sum competitors' capacity b-spline 1	7.74 (0.124)	5.80 (0.714)	7.26 (0.927)	7.094 (0.282)
Sum competitors' capacity b-spline 2	7.70 (0.123)	5.67 (0.715)	7.06 (0.929)	6.96 (0.326)
Sum competitors' capacity b-spline 3	7.76 (0.120)	5.80 (0.711)	7.17 (0.936)	7.50 (0.303)
Sum competitors' capacity b-spline 4	7.64 (0.127)	5.65 (0.719)	6.60 (0.964)	6.50 (0.334)
Sum competitors' capacity b-spline 5	7.88 (0.124)	5.96 (0.701)	6.82 (0.987)	7.31 (0.340)
Sum competitors' capacity b-spline 6	7.59 (0.124)	5.52 (0.746)	6.36 (0.992)	6.71 (0.391)
Own capacity b-spline 1	-2.24 (0.121)	-2.24 (0.121)	-2.15 (0.124)	-0.912 (0.301)
Own capacity b-spline 2	-1.36 (0.118)	-1.36 (0.118)	-1.31 (0.124)	-0.136 (0.308)
Own capacity b-spline 3	-0.752 (0.128)	-0.753 (0.128)	-0.702 (0.130)	-0.762 (0.354)
Own capacity b-spline 4	-0.182 (0.124)	-0.186 (0.125)	-0.120 (0.134)	1.27 (0.432)
Own capacity b-spline 5	0.074 (0.179)	0.0096 (0.178)	0.063 (0.181)	-0.831 (0.767)
Population b-spline 1		1.43 (0.892)	0.482 (2.30)	
Population b-spline 2		2.08 (0.679)	0.483 (0.876)	
Population b-spline 3		1.95 (0.727)	0.656 (1.04)	
Population b-spline 4		1.98 (0.666)	0.015 (0.802)	
Population b-spline 5		2.37 (0.881)	-0.566 (1.21)	
Capacity is per capita	No	No	No	Yes
Region fixed effects	No	No	Yes	No
Adjusted R^2	0.9994	0.9994	0.9995	0.9958
Estimated σ^2	0.0244	0.0242	0.0235	0.184

^aDependent variable is log of capacity level after adjustment. Number of capacity changes = 774. Parameters esti-

Table VII: investment target

7.5 Table IX: dynamic structural parameters

Read:

- Table IX reports recovered structural parameters before and after 1990.
- Entry costs rise after 1990. The mean entry cost rises from about 182,585 to about 223,326.
- Investment and divestment fixed costs also differ across regimes, but the clearest policy-relevant shift is in sunk entry costs.
- The difference column summarizes changes between pre- and post-Amendment regimes.

Economic message: Table IX is the paper's central structural result: the Clean Air Act Amendments raised the cost of entering the cement industry.

TABLE IX
DYNAMIC PARAMETERS^a

	Before 1990		After 1990		Difference	
	Mean	SE	Mean	SE	Mean	SE
Parameter						
Investment cost	230	85	238	51	-8	19
Investment cost squared	0	0	0	0	0	0
Divestment cost	-123	34	-282	56	-155	35
Divestment cost squared	3932	1166	5282	1130	1294	591
Investment Fixed Costs						
Mean (μ_i^+)	621	345	1253	722	653	477
Standard deviation (σ_i^+)	113	72	234	145	120	97
Divestment Fixed Costs						
Mean (μ_i^-)	297,609	84,155	307,385	62,351	12,665	34,694
Standard deviation (σ_i^-)	144,303	41,360	142,547	29,036	109	17,494
Scrap Values						
Mean (μ_ϕ)	-62,554	33,773	-53,344	28,093	9833	21,788
Standard deviation (σ_ϕ)	75,603	26,773	69,778	27,186	-6054	11,702
Entry Costs						
Mean (μ_κ)	182,585	36,888	223,326	45,910	43,654	21,243
Standard deviation (σ_κ)	101,867	22,845	97,395	14,102	-6401	12,916

^a Means of the parameters are reported for the pre-1990 period and the post-1990 period. Units are in thousands of dollars per ton for capital costs; the distributions are denominated in thousands of dollars. Standard errors were calculated via subsampling.

7.6 Table X: counterfactual policy experiments

Read:

- Table X compares post-1990 high-entry-cost equilibrium outcomes with a counterfactual that uses low pre-1990 entry costs.
- The upper panel simulates a new market with no incumbents and four potential entrants.
- The lower panel simulates a mature market with two incumbents and two potential entrants.
- With high entry costs, markets have fewer active firms, higher prices, lower quantities, and lower consumer surplus.
- Incumbent producer profits can rise because higher entry costs protect incumbents from competition.

Economic message: The counterfactuals show why static regulation cost analysis is misleading. Regulation can hurt consumers by reducing entry while helping incumbents by protecting them from competition.

TABLE X COUNTERFACTUAL POLICY EXPERIMENTS ^a						
	Low Entry Costs (Pre-1990)		High Entry Costs (Post-1990)		Difference	
	Mean	Standard Error	Mean	Standard Error	Mean	Standard Error
e Novo Market						
Total producer profit (\$ in NPV)	43,936.11	(7796.98)	33,356.87	(7767.22)	-11,182.04	(7885.20)
Profit firm 1 (\$ in NPV)	45,126.30	(10,304.87)	34,321.61	(9520.95)	-11,965.22	(11,684.96)
Total net consumer surplus (\$ in NPV)	1,928,985.09	(62,750.34)	1,848,872.52	(75,729.17)	-66,337.44	(58,404.32)
Total welfare (\$ in NPV)	2,116,810.12	(74,265.74)	1,992,937.65	(96,634.85)	-119,771.39	(49,423.06)
Periods with no firms (periods)	1.29	(0.08)	1.32	(0.09)	0.04	(0.08)
Periods with one firm (periods)	1.51	(0.37)	2.60	(0.86)	1.05	(0.78)
Periods with two firms (periods)	8.17	(4.68)	21.43	(9.92)	12.26	(9.99)
Periods with three firms (periods)	54.71	(20.22)	91.35	(21.27)	33.38	(18.85)
Periods with four firms (periods)	135.91	(24.64)	84.03	(32.67)	-46.73	(25.04)
Average size of active firm (tons)	980.71	(76.18)	1054.65	(85.17)	73.42	(74.01)
Average market capacity (tons)	3467.85	(108.21)	3352.23	(208.94)	-112.75	(107.84)
Average market quantity (tons)	3094.23	(161.57)	2987.61	(177.58)	-105.69	(89.41)
Average market price	66.66	(1.90)	68.12	(2.11)	1.47	(1.14)

(Continued)

	Low Entry Costs (Pre-1990)		High Entry Costs (Post-1990)		Difference	
	Mean	Standard Error	Mean	Standard Error	Mean	Standard Error
Mature Market						
Total producer profit (\$ in NPV)	223,292.75	(4831.95)	231,568.23	(5830.42)	9551.01	(5465.77)
Profit firm 1 (\$ in NPV)	549,179.30	(14,138.37)	579,742.32	(20,446.75)	32,968.00	(19,161.33)
Total net consumer surplus (\$ in NPV)	2,291,584.08	(52,663.88)	2,208,573.20	(62,906.14)	-62,974.37	(32,662.05)
Total welfare (\$ in NPV)	3,178,504.60	(60,267.34)	3,141,916.43	(62,618.02)	-30,999.56	(18,078.21)
Periods with no firms (periods)	0.00	(0.00)	0.00	(0.00)	0.00	(0.00)
Periods with one firm (periods)	0.00	(0.00)	0.00	(0.00)	0.00	(0.00)
Periods with two firms (periods)	8.63	(3.57)	23.20	(10.05)	14.13	(10.00)
Periods with three firms (periods)	61.32	(16.85)	98.37	(21.49)	35.73	(20.16)
Periods with four firms (periods)	131.52	(20.10)	78.38	(21.99)	-50.00	(27.59)
Average size of active firm (tons)	989.33	(44.45)	1059.31	(63.41)	73.48	(54.64)
Average market capacity (tons)	3502.49	(171.20)	3371.03	(191.87)	-117.56	(73.19)
Average market quantity (tons)	3123.42	(150.66)	3001.98	(165.51)	-108.16	(69.48)
Average market price	66.82	(1.64)	68.36	(1.91)	1.44	(0.85)

^a Industry distributions were simulated along 25,000 paths of length 200 each. All values are present values denominated in thousands of dollars. The new market initially has no firms and four potential entrants. The incumbent market starts with one 750,000 TPY incumbent and one 1.5M TPY incumbent and two potential entrants. Counts of active firms may not sum to 200 due to rounding off. Means and standard deviations were calculated by subsampling.

^b Where NPV denotes Net Present Value.

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Table X: counterfactuals, new market

Table X continued: counterfactuals, mature market

8 Synthesis

8.1 Why sunk entry costs are the key channel

The Portland cement industry has high fixed costs, long-lived capacity, and local market power. When regulation raises sunk entry costs, the immediate engineering cost is only part of the effect. Future entrants become less likely to enter, incumbents face less competition, and market prices rise. These entry dynamics create a consumer surplus loss that is absent from static engineering calculations.

8.2 Why incumbent welfare can move opposite to consumer welfare

If regulation raises entrants' costs more than incumbents' ongoing costs, incumbents may benefit from regulation. Entry barriers increase, future competition declines, and incumbent profits can rise. This is why a static analysis can get the sign of incumbent welfare effects wrong.

8.3 Why a dynamic model is necessary

The state variable, capacity, evolves slowly. Current regulation changes future market states by changing entry, exit, and investment. Product-market surplus depends on these future states, so a model that fixes the number of firms and capacities cannot evaluate the policy correctly.

9 Short summary

- The paper estimates the costs of the 1990 Clean Air Act Amendments in the Portland cement industry.
- It argues that engineering compliance costs miss dynamic market-structure effects.
- The model is a dynamic oligopoly with capacity, entry, exit, and investment decisions.
- Firms compete in quantities in regional cement markets.
- The paper uses the Bajari, Benkard, and Levin two-step estimator.
- The key structural finding is that the Amendments raised sunk entry costs.
- Higher entry costs reduce future entry and competition.
- Counterfactual simulations show large consumer surplus losses and possible incumbent profit gains.
- The full welfare analysis excludes benefits from cleaner air, so the paper measures only product-market costs.
- The broader lesson is that regulation in concentrated industries must be evaluated using dynamic equilibrium tools.

10 Conclusion

10.1 Core conclusion

Ryan shows that environmental regulation can have large indirect product-market costs through entry and market structure. In the Portland cement industry, the 1990 Clean Air Act Amendments increased sunk entry costs, leading to fewer entrants, higher prices, lower quantities, and lower consumer surplus.

10.2 Economic intuition

Regulation affects firms not only by requiring compliance expenditures but also by changing the incentives of potential entrants. In concentrated industries with high sunk costs, even a moderate increase in entry

costs can have persistent effects on market structure. The resulting reduction in competition is costly for consumers and can benefit incumbent producers.

10.3 Takeaway

The paper's central message is that policy analysis in concentrated industries must be dynamic. A static engineering estimate misses how firms respond through entry, exit, and investment, and therefore misses a central component of welfare.